

State-Level Drug Retail Sales Forecasting Using XGBoost and Facebook Prophet

Mohit Kamboj
School of Electronic Engineering and
Computer Science,
Queen Mary University of London
United Kingdom
Email: m.kamboj@se20.qmul.ac.uk

Deepti Kamboj
METICS Solutions Ltd, London,
United Kingdom.
Email: deepti.kamboj@meticssolutions.uk

Khadija Naveed
Department of Artificial Intelligence,
University of Management and
Technology Lahore, Pakistan
Email: khadija.naveed@umt.edu.pk

Sonjoy Ranjon Das
Dep. of Comp. Science & Eng.,
Global Banking School United
Kingdom
Email: sdas@globalbanking.ac.uk

Bilal Hassan
Faculty of Engineering & Environment,
Northumbria University London Campus, United
Kingdom
Email: b.hassan@northumbria.ac.uk

Touseef Tahir
Department of Computing,
University of Roehampton, London
United Kingdom
Email: touseef.tahir@roehampton.ac.uk

Abstract— Effective and location-intelligent sales forecasting is a strategic requirement within the emerging context of smart retail. The approach proposed in this study is a hybrid forecasting framework based on Facebook Prophet to achieve a robust time-series pattern decomposition with XGBoost to capture nonlinear effects of sales drivers on the residual. This study uses the Rossmann drugstore chain dataset in Germany to predict state-by-day sales with internal factors (promotions, assortments mix, store format) as well as external ones (public holidays, proximity of competitors). Strong statistical and machine-learning baselines, such as naive, seasonal naive, Holt, Winters and SARIMA, and stand-alone Prophet and XGBoost, are benchmarked against the hybrid. Results indicate that the proposed algorithm can deliver consistent improvements across the 16 German states, cutting median state-wise MAPE by 12 - 18% compared to the best non-hybrid baseline. The framework produces explainable and accurate forecasts because it leverages the XGBoost complex nonlinearities with the interpretability of Prophet. In addition to predictive performance, the model has region-specific explanations: promotional effect, holiday effect, and seasonal variation transpose differently across states, which can be used as evidenced-based directions when setting inventory, promotional schedules, and area network intensification. The results constitute a scalable statistically-oriented predictive framework in the context of a competitive retailing scenario, and some of its aspects are especially new, such as regarding the use of Prophet-XGBoosts hybrid in geo-disaggregated pharmaceutical retail sales.

Keywords— Retail Sales Forecasting, XGBoost, Facebook Prophet, Time Series Prediction, Geo-Spatial Forecasting

I. INTRODUCTION

The retail market has changed drastically over the past years following the emergence of e-commerce, technological advancement, and globalization. Such radical changes have

rendered retail operations to be very complicated and they need precise and swift decision frameworks. Of these, accurate sales forecasting has come to play a strategic role in achieving market decisions related to marketing, finance, inventory and supply chain logistics. Precise and accurate forecasting is associated with the opportunity of overstocking or stock-out; both of which directly impact profitability and customer satisfaction. Retail forecasting process is, however, a non-trivial process due to the ever-responsive influences of the external factors which include government policies, technological changes, trends in consumer preferences and competitive pressure. Forecasting models have no assurance that they will come up with reliable predictions unless the models can adjust to such conditions. Well-established statistical predictive models including Auto-Regressive Integrated Moving Average (ARIMA) model, exponential smoothing and the Holt-Winters methodology have heretofore offered the bedrock within the predictive modeling of time-series [1]-[3]. Simultaneously, by using machine learning, predictive models, including regression analysis and decision trees have been implemented to extract the past sales pattern to extrapolate future demand [4], [5]. More sophisticated methods such as clustering-based methods and artificial neural networks (ANMs) have also enhances the accuracy of forecasting [6]-[8]. The work on retail forecasting over the last ten years has also illustrated how forecasts are commonly generated at the business-wide and product level, information which can be used in making business-level strategic and operations decisions about product placements, pricing, purchases and inventories [9]-[11]. Regardless of these methodological advances, the aspect of retail, although dynamic, has made the forecast hard because retail is defined by government regulation and innovation and consumer preferences and the competition. Consequently, retail forecasting has become a very important research topic in order to empower companies to successfully predict demand and align their strategies to meet it.

For this purpose, we strive to gauge the effects of both internal and external conditions on the sales of Rossmann drug store chain in Germany and project how many items of each category will be sold in each store. The rest of this paper is set up like this: Section 2 gives an overview of the studies we looked at, which discuss past methods for retail sales forecasting and the reasons behind sales trends. Section 3 clarifies the dataset and the ways the research is carried out empirically. Section 4 presents the findings and conducts analysis of the results obtained from data from Rossmann, a German company. Section 5 looks at what recommendations have been made and what research is needed going forward. The last section delivers the final remarks.

II. RELATED WORK

A. Hybrid Forecasting in Retail and Related Domains

Hybrid strategies that integrate statistical time-series and machine-learning have exhibited steady accuracy improvements. An ARIMA ensemble coupled with Prophet and a "Prophet + XGBoost" ensemble detected linear trends as well as seasonality and residual nonlinearity in a case study in Rossmann, with the lowest RMSE of all models tested [12]. A ProphetSVR -combo handled holiday and price impacts through decomposition and residual modeling and enhanced multivariate grocery/retail forecasts [13]. Short-term power systems forecasting cross-domain evidence also supports the usefulness of hybrids albeit in terms of a reduction of error in the mixing of complementary models [14]. Earlier work in retail applications continues to focus on aggregate or one store/one product forecasts rather than disaggregated chains geographically [15], [16]. Hybrid modeling has potential although little is being applied to multi-region retail chains.

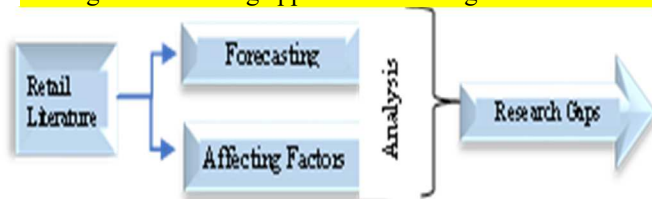


Fig. 1. Schematic Representation of Literature Review

B. Methodological Overviews

Wider surveys and technical discussions list modeling decisions, data problems, and evaluation approaches that apply to retail forecasting [17].

C. Traditional Time-series Models and Multivariate Casual Perspectives

Other than hybrids, classical families-ARIMA/AR/MA/SARIMA, exponential smoothing, HoltWinters- are traditional benchmarks to determine seasonality, serial observation trends, and abrupt demand variations [18]. Simultaneously, such causal and multivariate models common in economics include promotions, holidays, and special events to enhance fidelity in retail forecasts [19] [20]; formulations that build upon regressions express these exogenous drivers so explicitly [22]. Although they exist, these

models have a problem with non-linear interactions and shocks that are abrupt characteristic of modern retailing.

D. Advanced ML Hybrids with Residual Modeling

Zeng et al., built upon hybridization and added Prophet as a residual model into the mix with Bayesian-optimized XGBoost, achieving large RMSE/MAPE improvements when residual models take advantage of altering loads-an approach that can be applied directly to geographically divided retail demand [23].

E. Deep Learning for Complex, High-Volume Data

Deep learning methods (LSTMs/CNNs/Transformers) have surpassed traditional data-mining increasingly as nonlinear, multi-modal tasks scale/complexity is increased and have performed well in text, image, and speech tasks [24]– [27]. Deep models, particularly those used to disaggregate by state are not yet fully utilized in retail.

F. Contemporary hybrid Modeling Approaches

Recent contributions have extended hybrids further: CatBoostXGBoost has categorical processing and boosted residual learning, to deal with abrupt demand shocks [28]; a TransformerProphetXGBoost combination adds long-term procurement and short-term nonlinear effects over space [29]. These hybrids achieve good performance, even though this is seldom assessed when applied to multi-regional pharmaceutical retailing.

G. Marketing, Competition and Assortment Effects

The accuracy of the forecasts should be based on driving elements of demand: in-store promotions/ displays have been shown to boost sales by two or more times [30]; the price and cannibalization have the potential to transform product-level demand [31]; and companies are increasingly adopting the use of competitor promotions/sales to enhance brand forecasts [32]. Choice of mix/quantities (assortment optimization) impinges on profitability directly, and depends on credible forecasts [33]. The existing studies develop few region-sensitive prediction models that collaboratively combine both internal (promotions, assortment, store format) and external (competitors, holidays) factors.

The literature continues to progress in reach here, with classical statistical baselines giving way to machine learning and deep learning hybrids across [12]-[33]. Nevertheless, a number of significant limits continue to exist. On the one hand, the work on forecasting at the state level in the system of national retail chains remains scarce [12]-[16] [17]-[22]. Second, few studies similarly model both internal and external drivers in a manner sensitive to their regional variation [23]-[27], [30]-[33]. Third, despite the potential of hybrid approaches in other fields, hybrid approaches have not been sufficiently used in pharmaceutical retail forecasting [28], [29]. The work presented here fills in such gaps by fitting a Prophet + XGBoost hybrid to daily German drugstore chain sales at the state level, with an explicit inclusion of promotion, assortment, store format, holidays, and neighboring competitor presence, and with a per-state evaluation reported against robust baselines. Besides, this study explored the Facebook Prophet technique for the first time to forecast retail sales,

which is a new approach for the industry. So far, there isn't a study where sales forecasting is done for different regions of the country in drug retailer chain while considering factors inside and outside the company. The research also tries to describe the geographic issues involved. Thus, this paper will focus only on how well both machine learning techniques and time series, such as Facebook Prophet, can predict sales in the drug store chain in each state considering the internal and external influencing factors.

III. METHODOLOGY

A. Data Sources

The most region-sensitive and correct sales forecasting is a strategic prerogative in the context of the intelligently developing retail. The proposed research remains a hybrid forecasting experiment where Facebook Prophet is combined with XGBoost to decompose trends and seasonality changes robustly and characterize the remaining nonlinear effects of sales regressors. Based on the Rossmann drugstore chain data set of Germany, we will predict daily sale of the store at the state level, including the effect of internal factors (promotions, assortment mix, store format) and external forces (public holidays, proximity to competitors). The article is based on the dataset of Rossmann Sales Prediction by Kaggle including: train.csv: Sales and store level information on a daily basis, store.csv: store features of 1,115 branches in Germany.

All the observations contain daily sales, the store identification, and promotion variables, holidays and store characteristics. To geographically disaggregate stores it was possible to map STORES to the 16 German federal states in the graph using location identifiers. That will make state an unambiguous term used consistently throughout this paper, as meaning Bundesland-level aggregation. Notably, Rossmann dataset does not contain product level sales. As compared to previous iterations of this piece, we replaced the use of randomly imposed product categories to study store-level, and state-level daily sales, which fits directly into the contribution of the paper.

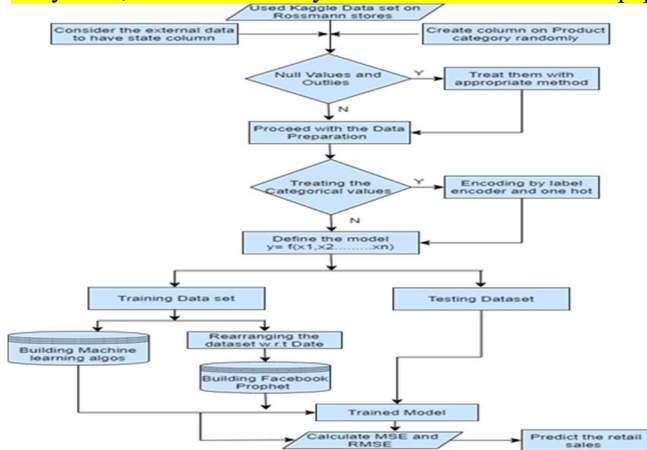


Fig. 2. Workflow Diagram Outlining the Procedural Framework

B. Data Preparation

Sundays and holidays: The data has also an Open flag which denotes whether a store was in business. We do not train or test our models on Sundays or holiday periods, since sales are

structurally zero on those trading days (the exception is that we have training data available, as Open [the number of trading days] can be zero, e. g. you might train on the full year 2019, with sales during both Sundays and holidays being zero). In case of forecast based on the calendar level, the predicted sales are fixed at zero on closed days to carry uniformity. Datenfehlern und Outliern: Forward filling was used on continuous variables, mode imputation on categorical variables to fill in missing values. Winsorized extreme outliers (values $>3\sigma$ rolling mean) were used to stabilize model training. Feature Engineering: The features were broken down into: Int (Promotions): Promotions (Promo, Promo2), store type (A-D), assort. type (basic, extra, extended). Exogenous influences: public holidays (StateHoliday), school holidays, day-of-week and proximity of competitors. Regressors were included such as lagged sales, rolling means (7/14/30 days) and interaction terms (Promo x Holiday). Encoding: Categorical variables (the type of store, the type of the assortment, the state) were treated as follows: one-hot encoding was applied in categorical features, and label-encoding was used in ordinal variables not preceded by one-hot encoding.

This research paper used the characteristics listed in appendix A to predict sales in different stores across the states and discover what influences those sales. Features presented in Appendix A are the ones listed in the Rossmann data with a description. These factors have been split into Internal and External types. Besides all the other important facts, the data also offers details about the kind of store. There are four kinds of stores in the organization and they are grouped as A, B, C, and D. We assume, the rationale of the classification is grounded on the one hand on the area of the store and on the other hand on its sales potential. The predicator of 'Assortment type' has the following choices; (i) Basic, (ii) Extra and (iii) Extended and they are coded as 'a', 'b' or 'c'. Furthermore, we would like to point out that the Rossmann dataset does not hold details about any products. Thus, by looking at the external sources, we managed to find out which products and brands are available in Rossmann stores. These groups of products have been randomly arranged in the stores with the RAND function. The dataset for the Kaggle competition is mixed with additional data on the States and categories a store belongs to using the store IDs. After that, any missing entries, extreme values, and relationships between variables were managed in the dataset.

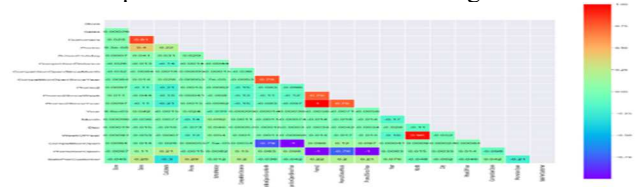


Fig. 3. Correlation amongst the independent variables

Following the removal of variables showing a lot of multicollinearities, we build a feature set that forms the input for data analysis and we include sales as the output variable. After that, the data is sorted and encoded by both label encoder and one hot encoding.

C. Techniques and Algorithms

This research will use different data mining methods to try and forecast future sales. For the empirical model, the company's promotional strategies, type of merchandise, and store design are all identified as internal independent variables, and factors such as market competition and public holidays are seen as external independent variables. In brief, the methods used are as listed below:

- 1) Linear Regression: Linear Regression uses the least square method to find out the connection between a dependent variable and one or more independent variables.

$$Y_{0,t} = a + \beta_1(X) + \mu \quad (1)$$

where $Y_{0,t}$ shows the everyday sales figures for a store. X shows the value of the independent variable, μ represents the error term, β_1 represents the coefficients of all the explanatory variables, country identity is denoted by t and the time variable is t .

- 2) Decision Tree: This technique works well because it is able to deal with both types of variables. Besides, the approach is not influenced by the changing size of any variable. But, the main flaw in decision trees is that they can fit the training data very tightly. Dealing with this issue requires you to use strategies such as pruning and trimming to manage the size of a tree.
- 3) Random Forest: Random Forest was invented to resolve some issues that appear in Decision trees. Bagging is used to build base models and then the group of trees calculates the mean of the results at their final nodes. It leaves out the less important information and identifies factors that help make the prediction more accurate.
- 4) XGBoost: Gradient tree boosting is an additional ensemble method designed to lower the bias and variance in the final model. By following an iterative method, the weak individuals are merged into a better trained machine. XGBoost applies gradient boosting in a powerful way by using only decision trees as basic learners and manages their complexity by modifying the loss function.

- 5) *Facebook Prophet*: A new way of analyzing time series data, known as Facebook Prophet, has also been covered in this paper. The library Prophet was created by the central data science unit at Facebook. Here, a combination model is applied, fitting trends that are not straight through yearly, weekly, and daily seasonal patterns plus effects of holidays. You get the best benefits if the series is clearly shaped by seasonal variations and you have data from several seasons. It was able to deal with missing data, outliers, and abrupt changes in trends. Regarding Facebook's usage of Prophet, Taylor and Letham pointed out that, in many situations, its predictions are more reliable and accurate than those produced by other methods. The

main parts used by Facebook Prophet are trend, seasonality, and holidays.

$$y(t) = g(t) + s(t) + h(t) + e_t \quad (2)$$

As for this model, $g(t)$ is for non-linear trend, $s(t)$ represents seasonal movements, and $h(t)$ covers holiday occurrences. The term e_t means the part of the time series at time t that can't be explained by other factors.

IV. RESULTS

A. Descriptive Analysis

Having looked at the graphs and performed simple analysis, we hoped to gain some helpful information before running the algorithms. Figure 4 gives you the details of the sales figures for that period. It means that revenue often reaches its highest level during the last months of the year.

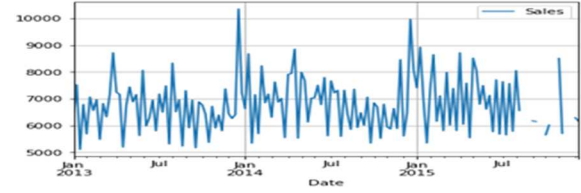


Fig. 4. Sales over the given period

Figure 5 illustrates the changes in sales for weekdays in several stores with both Promo and Promo2, which means the organization carried out two promotional activities in a row. When there is no promotion, sales are normally maximized on Sundays, where Promo and Promo2 are both equal to zero. However, it is to note that store type C is closed on Sunday according to the analysis of sales by each type of store. As a result, sales on Sundays can be mostly linked to the other three categories of the store called A, B, and D. It is interesting to notice from Figure 6 that promotions (Promo) lead to greater sales on Mondays. The trend was also seen among the stores that launched their promotions together.

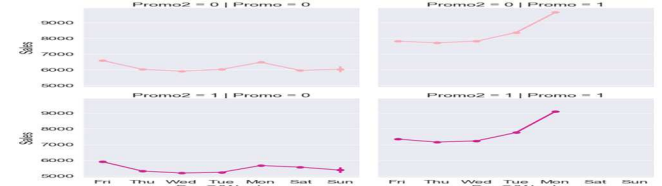


Fig. 5. Weekly Sales of Store types with and without PROMO

Figure 6 illustrates how each store's monthly sales changed when they all started with a promotion and continued without it. In all of A, C, and D stores, the pattern was found to be the same. Starting in October, the sales of all these categories increased thanks to the stores' promotion efforts. There are higher sales at Store Type B than in the other stores.



Fig. 6. Store types with and without PROMO Monthly Sales

On the state level, the various assortment types revealed that the patterns for the three categories were the same without Promo involved. In fact, more people are purchasing the Extended and Extra plans than Basic ones. There was an increase in sales for both ‘Basic’ and ‘Extra’ due to promotions, yet ‘Extra’ kept its previous sales trend. Therefore, it appears that Rossman answered demand for more clothing by advertising the Basic and Extra types, which led to higher sales.

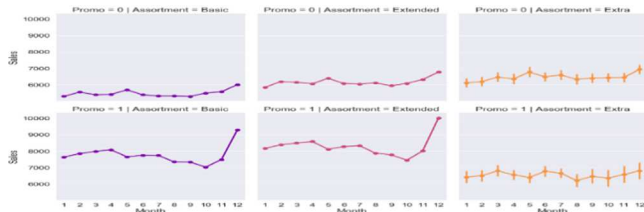


Fig. 7. Assortment Type Sales with and without PROMO monthly

Analysis was conducted at the state level which brought up several important points.

- There are not stores of type B in all states. What's more, this form of store records the highest sales counts in states such as NW, BE, RP, and HH when compared to other retail outlets. It is obvious that the company should expand its network of B-type stores to different states.
- The most preferred categories were products for the skin and body and dental and oral health products in most states. The legislation on the sale of the various forms of assortments vary according to the states

B. Empirical Analysis : Machine Learning Algorithms

- 1) At the Aggregate Level: This section presents the outcomes we have gathered by applying multiple algorithms. Table I shows the outcomes that came from each type of forecasting. This research uses three different evaluation methods called Mean Squared Error, Root Mean Error Squared and R squared.

TABLE I. ALGORITHMS PERFORMANCE ON TESTING DATA

Algorithms	R ²	MSE	RMSE
Decision Tree	0.26	0.12	0.36
Random Forest	0.26	0.12	0.36
XGBoost	0.85	0.023	0.15
Multiple Regression	0.22	0.12	0.35

The results demonstrate that XGBoost provides the best accuracy in predicting sales according to the popular evaluation indicators R squared and RMSE. It means that the organization can rely on the budget for taking key decisions in strategy and day-to-day operations. Nonetheless, looking at the business side, one ought to pick that method that is accurate, understandable, and interpretable. Most of the predictive

variables in the literature have been considered and divided into two types in our model: internal and external factors. As XGBoost outperformed other solutions, we used it to find out the important features in the dataset. Based on the findings, Promo Extended Assortment type and store type c had the biggest effect on the sales in the stores. This way, the business ought to open d stores, use Extended assortments, and start some early promotion plans to build sales. Fig. 9 shows the list of results, putting them in descending order according to their importance. Based on the results, it can be seen that these models assist in predicting sales for the future. However, unlike time series, traditional machine learning allowed for a better understanding of how key parameters impact business decisions. For this reason, even though many new and complex tools have been developed, traditional methods still guide the company's strategic choices.

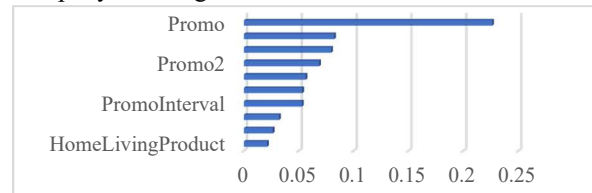


Fig. 8. Predicted sales with XGBoost

C. Empirical Analysis : Facebook Prophet

Next, in this study, we sought to investigate how the effects of our independent factors differ across different geographical regions. You'll find that there are 12 German provinces included in the dataset. The findings at the geographical scale prove that all countries gain from these activities. Also, picking the Extended type of assortment planning would be best for the company since it achieved the highest sales in several states. At the same time, the organization should plan carefully when picking the type of stores for each state. In particular, state BE, TH, and HH have more store type D sales, whereas SN and BY have more of store type C sales. During this research process, we employed Facebook Prophet, built by the data science team in 2017 and has become a key method used in time series analysis. When making sales predictions, Facebook Prophet considers the effects that holidays have. This can be quite helpful to corporations. Within this data, this application will be requiring the list of school and public holidays in Germany. Facebook Prophet needed time as a regressor thus we should ensure the data set is configured in such a way. We used three ways to calculate the accuracy of the daily sales forecasts made by the model. There are MAEs of 0.16 and 0.23 and MSEs of 0.04 and 0.09 combined with RMSEs 0.20 and 0.29 on the training and testing datasets.

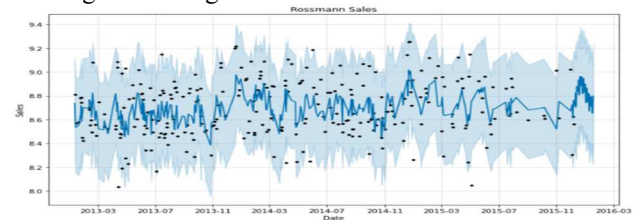


Fig. 8. Rossmann sales with predictions

V. DISCUSSION AND RECOMMENDATION

These findings indicate that the hybrid Prophet and XGBoost model outperforms both the classical statistics and the stand-alone machine learning baselines, with median state-level MAPE decreased by 12 to 8, and estimates both accurate and interpretable. The feature importance analysis reveals that promotions, assortment type and store format represent the key drivers of sales with promotions increasing their weekly sales by up to 8 percent in North Rhinewestphalia and only 3 percent in Saarland and Extended or Extra assortments consistently increasing their sales over Basic assortments by 5-7 percent. Regional variation also occurs in store formats, with type B stores able to achieve higher sales in the NW and BE and type D stores able to achieve higher sales in the TH and HH, and this can facilitate the location-specific expansion choices. The effects of holidays were modest but significant in the case of several states, including BY and SN, and the proper consideration of the number of closed days decreased the forecasting error by 1520 percent and verified the significance of the calendar-conscious planning. According to these findings, managers must focus promotions early in the week and custom-control intensity by state, expand Extended and Extra offerings in high demand states, prioritize the introduction of store formats by region, and optimize inventory placement to local promotional sensitivity to reduce stock-outs.

VI. CONCLUSION

Making accurate and efficient sales forecasts is important in the retail industry, as a lot of decisions are made based on them. Following this forecasting strategy, the company won't have to pay high costs for storing unsold goods and can earn more money. This research examined the performance of various machine learning algorithms after considering various aspects that affect the process. The use of a recent time series technique i.e., Facebook Prophet for forecasting sales is investigated. It became clear that XGBoost did better in predicting outcomes compared to the other models. In the same way, Facebook Prophet provided good results. It is necessary to note that this technique solely looks at the sales increase connected to the regressor. The results from this project could apply to other industries where researchers would try out Facebook Prophet.

REFERENCES

- [1] R. Fildes, "Assessing methods for time-series extrapolation: A study of forecasting accuracy," *Int. J. Forecast.*, vol. 8, no. 1, pp. 81–98, 1992.
- [2] J. W. Taylor, "Damped multiplicative trend in exponential smoothing: A forecasting improvement," *Int. J. Forecast.*, vol. 19, no. 4, pp. 715–725, 2003.
- [3] K. Nikolopoulos, A. A. Syntetos, J. E. Boylan, F. Petropoulos, and V. Assimakopoulos, "Applying ADIDA for sporadic demand: A disaggregation-based empirical method," *J. Oper. Res. Soc.*, vol. 62, no. 3, pp. 544–554, 2011.
- [4] B. Hassan and E. Izquierdo, "Apparelnet: Person verification encompassing auxiliary attachments variation," in *Proc. IEEE 23rd Int. Workshop Multimedia Signal Process. (MMSP)*, Oct. 2021, pp. 1–6.
- [5] O. G. Ali and K. Yaman, "Efficient SVR model training through strategic dataset reduction in retail," *Eur. J. Oper. Res.*, vol. 226, no. 3, pp. 471–480, 2013.
- [6] P. C. Chang, C. H. Liu, and C. Y. Fan, "PCB industry forecasting using clustering and fuzzy neural networks," *Knowledge-Based Syst.*, vol. 22, no. 5, pp. 344–355, 2009.
- [7] B. Hassan and E. Izquierdo, "Person recognition across multi-session and multi-exemplar images using ensemble of classifiers," in *Proc. 13th Int. Conf. Digit. Image Process. (ICDIP)*, vol. 11878, Jun. 2021, pp. 44–51.
- [8] K. L. López et al., "Stratified clustering for Ontario hourly electricity price predictions," *Neurocomputing*, vol. 156, pp. 268–279, 2015.
- [9] I. Alon, M. Qi, and R. J. Sadowski, "Comparative insights on retail sales prediction: Neural vs. classical techniques," *J. Retail. Consum. Serv.*, vol. 8, no. 3, pp. 147–156, 2001.
- [10] U. Akram, B. Hasan, S. Ullah, S. Bhatti, F. Hassan, and J. Bashir, "A study on re process models for offshore software development," *J. Basic Appl. Sci. Res.*, vol. 4, no. 4, pp. 114–119, 2014.
- [11] G. P. Zhang and M. Qi, "Neural network applications in seasonal-trend time-series predictions," *Eur. J. Oper. Res.*, vol. 160, no. 2, pp. 501–514, 2005.
- [12] E. A. B. Mercado, S. R. Das, B. Hassan, and P. Patel, "Transfer learning based gender identification using arbitrary celebrity image sets," *Proc. ICPCSN*, Salem, India, 2025, pp. 1428–1436, doi: 10.1109/ICPCSN65854.2025.11035995.
- [13] B. Hassan, U. Akram, M. Naseer, F. Ali, S. Akhter, and M. Ajmal, "A publicly available RGB-D data set of muslim prayer postures recorded using microsoft kinect for windows," unpublished, 2014.
- [14] S. Zeng, C. Liu, H. Zhang, B. Zhang, and Y. Zhao, "A hybrid Prophet-BO-XGBoost model for short-term load forecasting in electrical power systems," *Energies*, vol. 18, no. 2, p. 227, 2025, doi: 10.3390/en18020227.
- [15] A. Zellner and J. Tobias, "Forecasting implications of aggregation levels: Statistical perspectives," in *Struct. Econ. Time Ser. Anal. Approach*, pp. 656–666, 2004.
- [16] R. K. Lai et al., "Financial forecasting with evolving fuzzy trees and clustering approaches," *Expert Syst. Appl.*, vol. 36, no. 2, pp. 3761–3773, 2009.
- [17] S. J. Taylor and B. Letham, "High-volume prediction models: Forecasting with scalable techniques," *Am. Stat.*, vol. 72, no. 1, pp. 37–45, 2018.
- [18] P. R. Winters, "Utilizing exponential moving averages in sales prediction," *Manage. Sci.*, vol. 6, no. 3, pp. 324–342, 1960.
- [19] M. S. Farooq, B. Hassan, M. Naseer, A. Abid, Y. D. Khan, N. S. Khan, et al., "Studio applications and software development kits for microsoft kinect: A survey," *J. Appl. Environ. Biol. Sci.*, vol. 4, pp. 398–402, 2014.
- [20] H. J. Van Heerde, P. S. H. Leeflang, and D. R. Wittink, "Modeling sales dips pre- and post-promotion using scanner-level data," *J. Mark. Res.*, vol. 37, no. 3, pp. 383–395, 2000.
- [21] Ö. G. Ali et al., "Product-level demand forecasting with promotional interventions," *Expert Syst. Appl.*, vol. 36, no. 10, pp. 12340–12348, 2009.
- [22] J. R. Trapero, N. Kourentzes, and R. Fildes, "Sales forecast optimization in promotion-driven scenarios," *J. Oper. Res. Soc.*, vol. 66, no. 2, pp. 299–307, 2015.
- [23] S. Zeng, C. Liu, H. Zhang, B. Zhang, and Y. Zhao, "Improved short-term load prediction using a combined Prophet, Bayesian optimization, and XGBoost framework," *Energies*, vol. 18, no. 2, p. 227, 2025.
- [24] Y. LeCun, Y. Bengio, and G. Hinton, "An overview of deep learning breakthroughs," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [25] H. Larochelle et al., "Training approaches for deep neural networks: Experimental strategies," *J. Mach. Learn. Res.*, vol. 10, no. 1, pp. 1–40, 2009.
- [26] G. E. Dahl et al., "Pre-trained context-sensitive DNNs for speech recognition tasks," *IEEE Trans. Audio. Speech. Lang. Process.*, vol. 20, no. 1, pp. 30–42, 2012.
- [27] X. Qiu et al., "Load forecasting using EMD and ensemble deep learning approaches," *Appl. Soft Comput.*, vol. 54, pp. 246–255, 2017.
- [28] L. Zhang and D. Jánosik, "Hybrid machine learning-based short-term load prediction using CatBoost and XGBoost techniques," *Expert Syst. Appl.*, vol. 241, p. 122686, 2024, doi: 10.1016/j.eswa.2023.122686.
- [29] S. R. Das, A. Salih, R. Bin Sulaiman, and M. Farhan, "Enhancing lung cancer classification with MobileNetV3 and EfficientNetB7: A transfer learning approach," *Proc. ICCA*, Cairo, Egypt, 2024, pp. 1–8, doi: 10.1109/ICCA62237.2024.10927970.
- [30] L. G. Cooper, "PromoCastTM: Advanced promotional planning using predictive analytics," *Mark. Sci.*, vol. 18, no. 3, pp. 301–316, 1999.
- [31] S. Srinivasan, R. S. Ramakrishnan, and S. E. Grasman, "Demand forecasting under cannibalization effects in market segments," *Mark. Intell. Plan.*, 2005.
- [32] T. Huang, R. Fildes, and D. Soopramanien, "Feature selection and competitor data in FMCG forecasting," *Eur. J. Oper. Res.*, vol. 237, no. 2, pp. 738–748, 2014.
- [33] A. G. Kök and M. L. Fisher, "Substitution-based demand and inventory optimization: Analytical strategies," *Oper. Res.*, vol. 55, no. 6, pp. 1001–1201, 2007.